



ANOVA

HYPOTHESIS TESTING

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TOPIC OUTLINE

Bartlett's Test

One-Way ANOVA

Welch's ANOVA



BARTLETT'S TEST



BARTLETT'S TEST

Bartlett's Test is a statistical test used to assess whether multiple groups have equal variances.

Hypothesis

$$H_0: \sigma_1 = \sigma_2 = \sigma_3 = \dots \sigma_n$$

$$H_a: \text{at least one } \neq \text{ (p-value } < \alpha \text{)}$$

Assumptions

- Continuous data
- Normal data

syntax

```
from scipy import stats

b_stat, p_value = stats.bartlett(
    sample_1 data,
    sample_2 data,
    sample_3 data, ... sample_n data)
```



ONE-WAY ANOVA



ONE-WAY ANOVA

One-Way ANOVA is a statistical test used to compare the means of **three or more** independent groups to determine if there is a statistically significant difference among them.

Hypothesis

$$H_o: \mu_1 = \mu_2 = \mu_3 = \dots \mu_n$$

$$H_a: \text{at least one } \neq \text{ (p-value } < \alpha)$$

Assumptions

- Continuous data
- Normal data
- Equal variances

syntax

```
from scipy import stats

f_stat, p_value = stats.f_oneway(
    sample_1 data,
    sample_2 data,
    sample_3 data, ... sample_n data)
```



WELCH'S ANOVA



WELCH'S ANOVA

Welch's ANOVA is a variation of the classic one-way ANOVA that does **not** assume equal variances across groups.

Hypothesis

$$H_0: \mu_1 = \mu_2 = \mu_3 = \dots \mu_n$$

$$H_a: \text{at least one } \neq \text{ (p-value } < \alpha)$$

Assumptions

- Continuous data
- Normal data
- **Unequal** variances

syntax

```
import pingouin as pg

result = pg.welch_anova(
    dv = 'data_values column',
    between = 'group_column',
    data = df)

p_value = result['p-unc'].values[0]
```

note: stuck dataset



EXERCISE

The dataset contains power output measurements from three hydroelectric power plants. Verify the maintenance department claim that one of the plants is underperforming.

dataset

hydroelectric-sample.csv

Solution

Let $\alpha = 0.05$

Hypothesis

$H_o: \mu_1 = \mu_2 = \mu_3 = \dots \mu_n$

$H_a: \text{at least one } \neq \text{ (p-value } < \alpha)$



LABORATORY

